Bead-Spring Polymer Model: Molecular Dynamics, Langevin Dynamics & Scaling with Chain Length

Chinmoy Saikia
20231074

1 Introduction

A polymer in a good solvent acts like a self-avoiding chain. Its size follows a power law,

$$R_g \sim N^{\nu}, \tag{1}$$

where N is the number of monomers and $\nu \approx 0.588$ is the Flory exponent in three dimensions. The slowest internal motion of the chain relaxes over a time

$$\tau_R \sim N^{1+2\nu}, \tag{2}$$

which is $N^{2.176}$ for $\nu = 0.588$. This is the Rouse result with excluded volume and without hydrodynamic interactions. The results might differ slightly since we have used WCA potential to mimic the effect of excluded volume.

The report has three parts.

1. **Plain MD** : A chain of 100 beads is evolved with velocity Verlet. Total energy and momentum should be conserved, and we look at how well they are.
2. **Langevin MD** : A thermostat is added, and we measure R_g and τ_R for several values of N to find ν.
3. **Parallel averaging** : Single runs are noisy, so many independent runs are done for each N and averaged. We then report the scaling results (Section 6) and discuss finite-size effects.

2 Model

The polymer is a chain of N beads of mass m. Two potentials act on the beads.

Excluded volume. Every pair of beads repels through the Weeks-Chandler-Andersen (WCA) potential, which is the purely repulsive part of the Lennard-Jones potential:

$$U_{\mathrm{WCA}}(r) = \begin{cases} 4\epsilon\left[\left(\dfrac{\sigma}{r}\right)^{12} - \left(\dfrac{\sigma}{r}\right)^{6}\right] + \epsilon, & r < r_c = 2^{1/6}\sigma, \\ 0, & r \geq r_c. \end{cases} \tag{3}$$

The repulsion acts between all pairs, including neighbours along the chain.

Bonds. Neighbouring beads along the chain are joined by a harmonic spring,

$$U_{\text{spring}}(r) = \tfrac{1}{2}\kappa\,(r - r_0)^2. \tag{4}$$

The total potential energy is the sum of both terms over all pairs and bonds. All quantities are in reduced units, with $\sigma = \epsilon = m = k_B T = 1$. Table 1 lists the parameters.

Quantity	Symbol	Value
Bead mass	m	1
Thermal energy	$k_B T$	1
Bead size, WCA energy	$\sigma,\ \epsilon$	1, 1
Spring constant	κ	100
Spring rest length	r_0	1
WCA cutoff	r_c	$2^{1/6}$
Neighbour-list skin		0.3
Friction (Langevin runs)	γ	1
Time step (Langevin runs)	Δt	0.001

Table 1: Model and simulation parameters in reduced units.

Neighbour list. Checking all $N(N-1)/2$ pairs at every step is slow. Therefore we keep a list of pairs closer than $r_c + \text{skin}$. The list is rebuilt only when some bead has moved more than half the skin distance since the last build. The execution is analogous to the subroutine discussed in Computational Physics Molecualr Dynamics segment.

3 Molecular Dynamics with Velocity Verlet

3.1 Algorithm

Without a thermostat the beads follow Newton's equations, $m\ddot{\mathbf{r}}_i = \mathbf{F}_i$. Velocity Verlet advances positions and velocities by one time step Δt as follows:

$$\mathbf{v}_i\big(t + \tfrac{\Delta t}{2}\big) = \mathbf{v}_i(t) + \frac{\Delta t}{2m}\,\mathbf{F}_i(t), \tag{5}$$

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \Delta t\,\mathbf{v}_i\big(t + \tfrac{\Delta t}{2}\big)\,, \tag{6}$$

$$\mathbf{v}_i(t + \Delta t) = \mathbf{v}_i\big(t + \tfrac{\Delta t}{2}\big) + \frac{\Delta t}{2m}\,\mathbf{F}_i(t + \Delta t), \tag{7}$$

where $\mathbf{F}_i(t + \Delta t)$ is computed from the new positions between the second and third line. The steps of the program are:

1. Place the $N = 100$ beads in a chain and draw velocities from a Maxwell-Boltzmann distribution at $k_B T = 1$. Subtract the centre-of-mass velocity so the total momentum starts at zero.
2. Build the neighbour list and compute the initial forces.
3. Repeat for every time step: half velocity update, position update, neighbour-list check (rebuild if needed), new forces, second half velocity update.
4. At regular intervals write out the kinetic energy $K = \frac{m}{2}\sum_i |\mathbf{v}_i|^2$, the two potential energies, the total energy $E = K + U_{\text{WCA}} + U_{\text{spring}}$, and the total momentum $\mathbf{P} = m\sum_i \mathbf{v}_i$.

3.2 Energy conservation

Figure 1 shows the energies over a run of length 100. The kinetic energy and the two potential energies change a lot as the chain relaxes from its starting shape, and then fluctuate around steady values. Their sum E is a flat line on this scale.

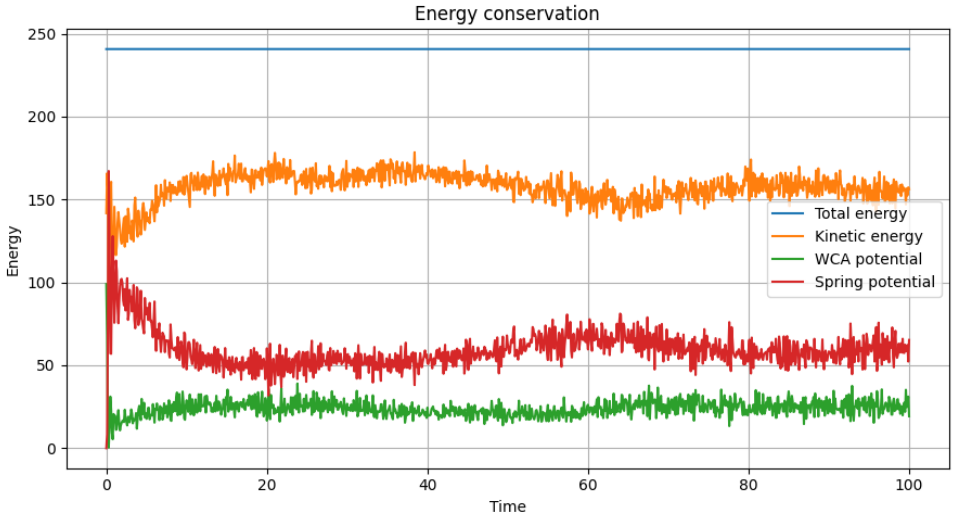


Figure 1: Kinetic, WCA, spring and total energy against time for the 100-bead chain (velocity Verlet). The parts exchange energy, but the total stays constant.

Figure 2 zooms in on the total energy. It is not perfectly constant. After a small jump in the first output interval, E stays inside a band of width about 0.006 around 240.74. In relative terms, $(E(t) - E_0)/|E_0|$ stays between about 1.5×10^{-5} and 4.2×10^{-5} for the rest of the run. The jump at the start is likely because the initial configuration has the largest forces (the WCA energy starts near 100 and drops within the first few steps), so the integration error is largest there.

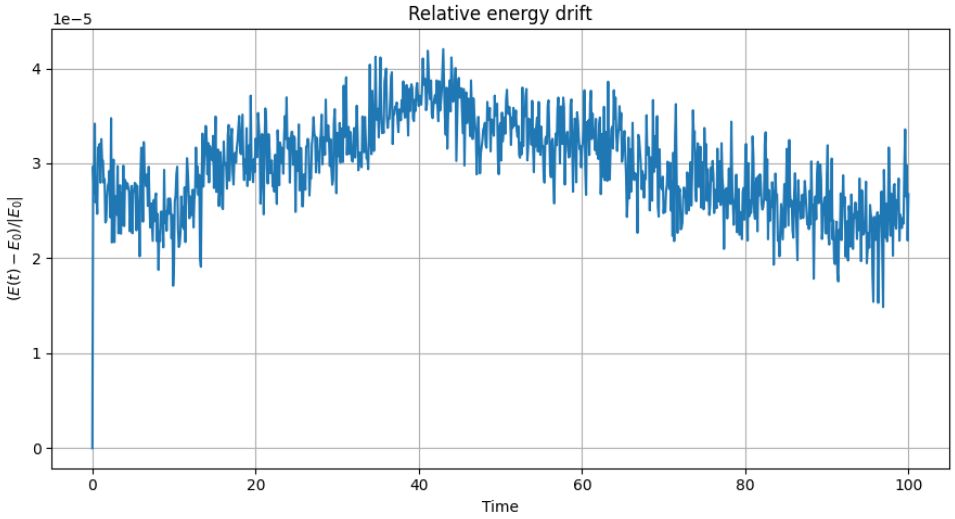


Figure 2: Relative deviation from the initial energy. The energy fluctuates by about 10^{-5} in relative terms and shows no steady drift.

There are two separate sources of error here, and they have very different sizes.

Finite time step. Velocity Verlet is time-reversible and symplectic. It does not conserve E

exactly, but it conserves a slightly modified energy $\tilde{E} = E + O(\Delta t^2)$. The true energy therefore oscillates around a constant value by an amount of order Δt^2, and it does not drift away over long times. This is the origin of the fluctuations in Figure 2, and it is why Verlet is preferred over more conventional integration schemes like RK4.

Round-off. In double precision, round-off errors are typically of order 10^{-16} per operation, which is much smaller than the integration error and is therefore not clearly visible in the energy fluctuations. Momentum provides a more sensitive test because the internal forces occur in equal and opposite pairs, so the total momentum

$$\mathbf{P} = \sum_i m\mathbf{v}_i$$

should remain exactly zero in exact arithmetic. Figure 3 shows that $|\mathbf{P}|$ remains extremely small, reaching only about 7×10^{-13} over the entire simulation. The small drift is therefore consistent with the accumulation of floating-point round-off errors over many time steps, rather than a physical loss of momentum conservation.

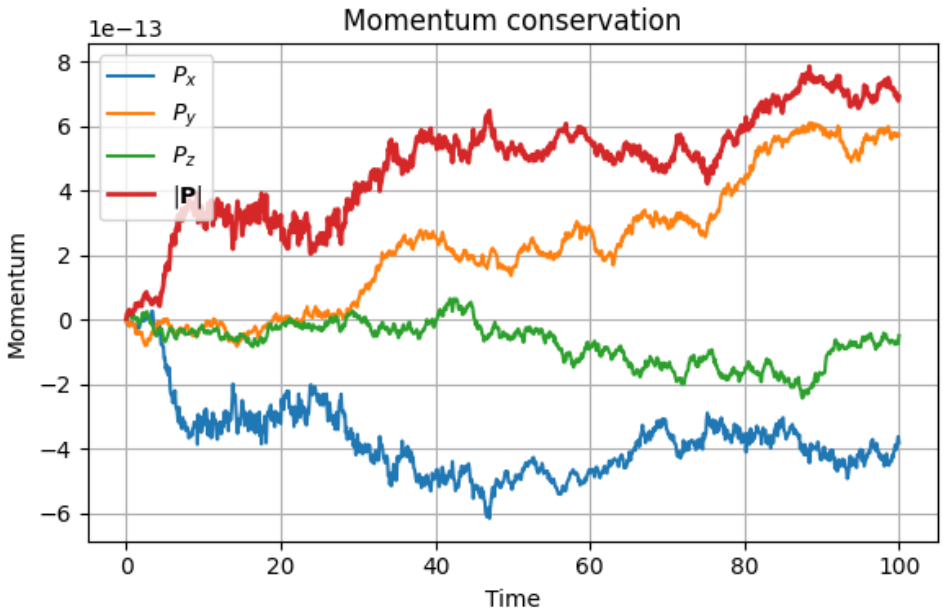


Figure 3: Total momentum components and magnitude against time. The values are of order 10^{-13}, which is round-off error.

4 Langevin Dynamics with the BAOAB Scheme

4.1 Equation of motion and integrator

Plain MD conserves energy, so the temperature is not fixed. To simulate a polymer in a heat bath we add friction and random kicks:

$$m\ddot{\mathbf{r}}_i = \mathbf{F}_i - \gamma\,\dot{\mathbf{r}}_i + \boldsymbol{\xi}_i(t), \qquad \langle \xi_{i\alpha}(t)\,\xi_{j\beta}(t')\rangle = 2\gamma k_B T\,\delta_{ij}\delta_{\alpha\beta}\,\delta(t-t'). \tag{8}$$

The friction removes energy and the noise adds it, and together they hold the temperature at $k_B T = 1$. We integrate with the BAOAB scheme, which splits one step into five parts:

$$\mathbf{B}\quad \mathbf{v} \leftarrow \mathbf{v} + \frac{\Delta t}{2m}\mathbf{F} \qquad \text{(half kick)}$$

$$\mathbf{A}\ \ \mathbf{r} \leftarrow \mathbf{r} + \frac{\Delta t}{2}\mathbf{v} \quad \text{(half drift)}$$

$$\mathbf{O}\ \ \mathbf{v} \leftarrow c_1\mathbf{v} + c_2\,\mathbf{R} \quad \text{(thermostat)}$$

$$\mathbf{A}\ \ \mathbf{r} \leftarrow \mathbf{r} + \frac{\Delta t}{2}\mathbf{v} \quad \text{(half drift)}$$

$$\mathbf{B}\ \text{recompute } \mathbf{F}, \text{ then } \mathbf{v} \leftarrow \mathbf{v} + \frac{\Delta t}{2m}\mathbf{F} \quad \text{(half kick)}$$

Here $\mathbf{R}$ is a vector of independent Gaussian random numbers with zero mean and unit variance, and

$$c_1 = e^{-\gamma \Delta t/m}, \qquad c_2 = \sqrt{\frac{k_B T}{m}\,(1 - c_1^2)}. \tag{9}$$

The Gaussian numbers are made from uniform random numbers with the Box-Muller transformation. Each chain starts as a self-avoiding random walk, built bead by bead with a new random direction whenever the subsequent bead overlaps an earlier one. This shape is already close to an equilibrium chain, so it needs less time to relax as compared to a straight chain, this is different from the initialization we adopted in simple MD since the constants need the polymer to equilibrate before they can be estimated.

4.2 A single run for $N = 100$

Figure 4 shows one run of length 1000 for $N = 100$, started from a straight chain. The temperature fluctuates around the target value 1, as it should. The energy and total momentum are no longer conserved, because the thermostat exchanges energy and momentum with the bath.

The radius of gyration starts near 29, which is the value for a straight chain of 100 beads, and decays to about 10. Averaging R_g over the second half of the run gives 9.94 with a standard deviation of 1.31. The figure also shows a limit of this run: R_g is still drifting down at $t = 1000$. The relaxation time of a 100-bead chain is about 900 to 1200 in these units (Section 6), so this run is only about one relaxation time long. The chain is not fully equilibrated. This is why the in the later runs we set length in units of τ_R instead of a fixed number of steps.

4.3 Quantities measured

Radius of gyration.

$$R_g^2 = \frac{1}{N}\sum_{i=1}^{N} |\mathbf{r}_i - \mathbf{r}_{\text{cm}}|^2\,. \tag{10}$$

We average R_g^2 over time and then take the square root, because the scaling law is for $\langle R_g^2 \rangle^{1/2}$.

Relaxation time. The slowest internal motion of the chain is the first Rouse mode,

$$\mathbf{X}_1(t) = \frac{1}{N}\sum_{i=1}^{N} \cos\left[\frac{\pi\,(i - \frac{1}{2})}{N}\right] \mathbf{r}_i(t). \tag{11}$$

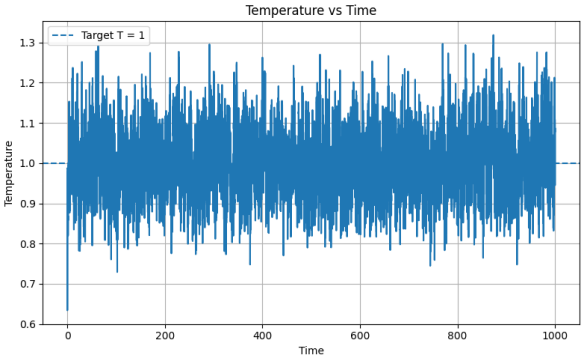
Its normalised autocorrelation function is

$$C(t) = \frac{\langle \delta\mathbf{X}_1(t_0 + t) \cdot \delta\mathbf{X}_1(t_0) \rangle}{\langle |\delta\mathbf{X}_1|^2 \rangle}, \tag{12}$$

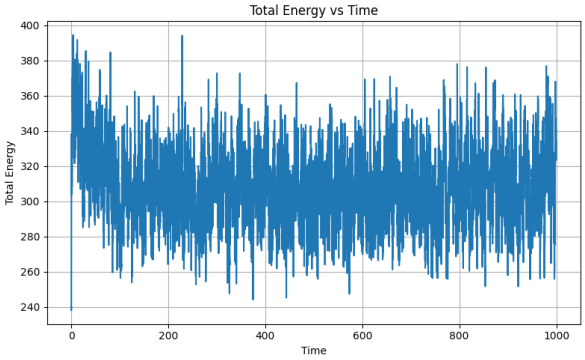
where $\delta\mathbf{X}_1$ is $\mathbf{X}_1$ minus its mean. If $C(t) \sim e^{-t/\tau_R}$, the relaxation time is the integrated autocorrelation time,

$$\tau_R = \Delta t_s \left[\frac{1}{2} + \sum_{k \geq 1} C(k\,\Delta t_s) \right], \tag{13}$$

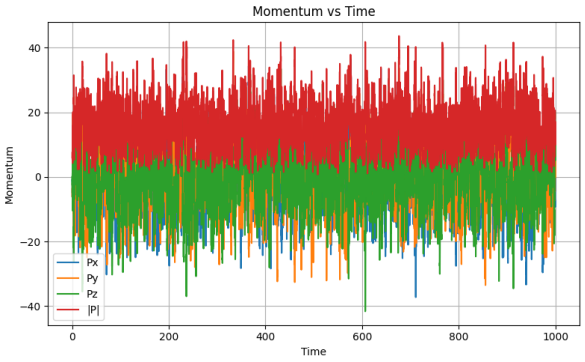
where Δt_s is the sampling interval.



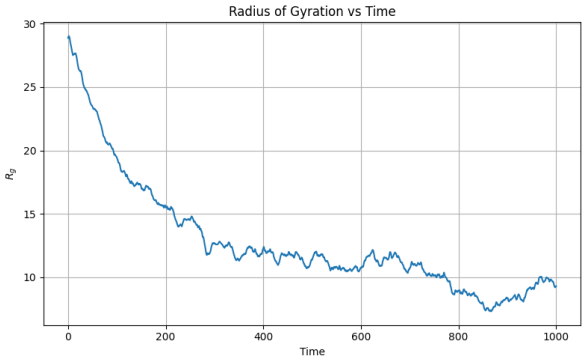
(a) Instantaneous temperature.



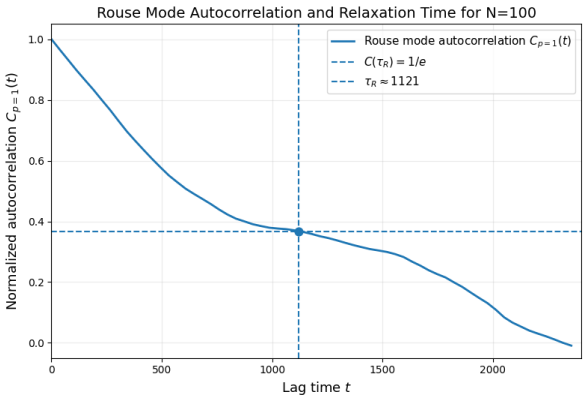
(b) Total Energy versus Time



(c) Momentum versus Time



(d) Radius of gyration stabilizing over time.



(e) Autocorrelation versus Lag time

Figure 4: Langevin run for $N = 100$. The temperature stays near the target value 1. R_g decays from about 29 towards its equilibrium value. Energy and momentum fluctuate about their mean due to the introduction of the thermostat. Autocorrelation decays to zero with increasing lag time.

The sum stops at the first lag where $C \leq 0$, because beyond that point it only adds noise. This cut makes τ_R slightly too small on average, which is one reason we average over many runs later.

For planning the run length we also use the estimate

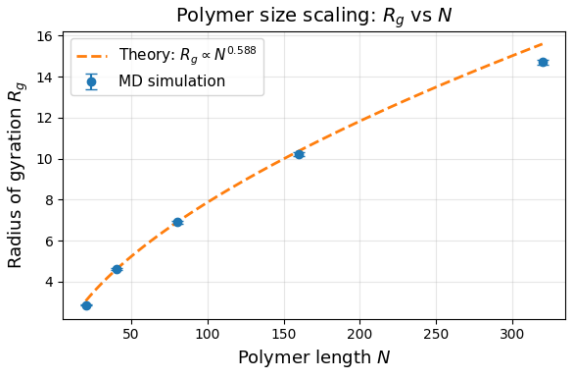
$$\tau_{\mathrm{est}} = \frac{\gamma\, N^{1+2\nu}}{3\pi^2}, \qquad \nu = 0.588. \tag{14}$$

4.4 First attempts with single runs

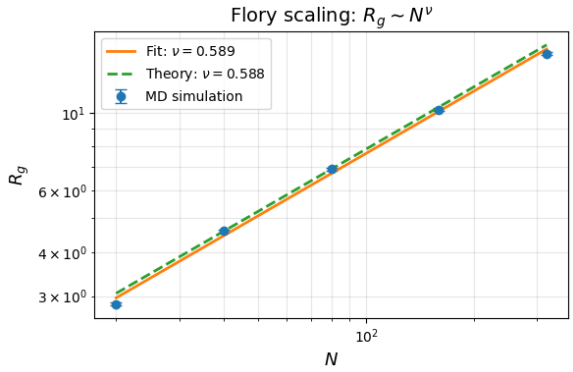
Our first scaling measurements used one run per chain length. We repeated them with different random seeds and different equilibration times. Table 2 lists the exponents from straight-line fits of $\log R_g$ and $\log \tau_R$ against $\log N$. The value of ν from τ_R is $(z-1)/2$, where z is the fitted slope of $\log \tau_R$.

Attempt	Setup (one run per N)	N range	ν from R_g	ν from τ_R
1	average over $T_{\mathrm{eq}} < t < 3T_{\mathrm{eq}}$, 5 sizes	20 to 320	0.589	not measured
2	fixed seed 69	10 to 300	0.627	0.577
3	seed from system clock	10 to 300	0.620	0.570
4	seed from system clock	10 to 300	0.653	0.645
5	equilibration raised to $5\,T_{\mathrm{eq}}$	10 to 180	0.618	0.605

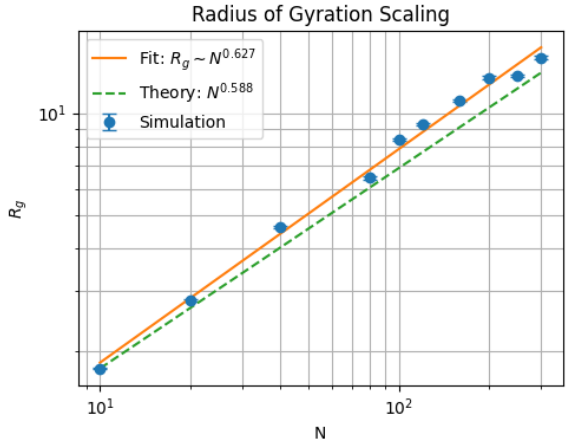
Table 2: Scaling exponents from single-run attempts.



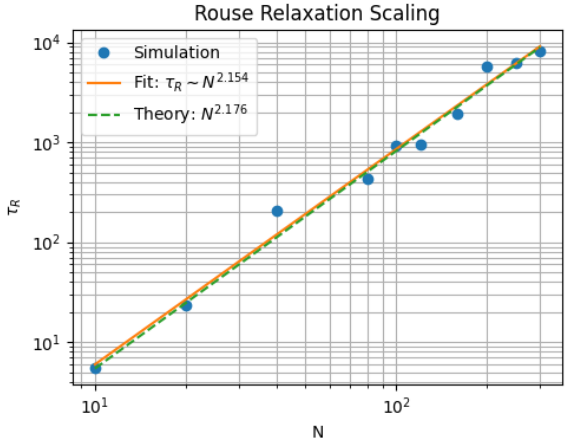
(a) Attempt 1.1



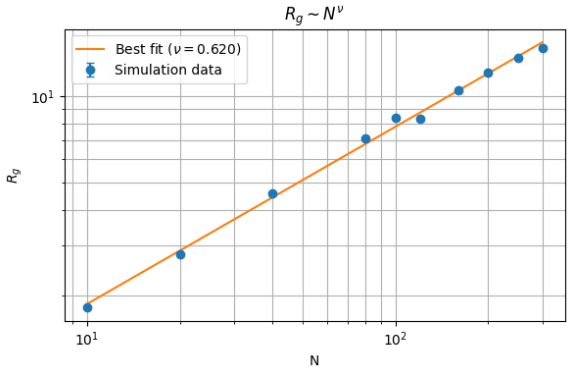
(b) Attempt 1.2



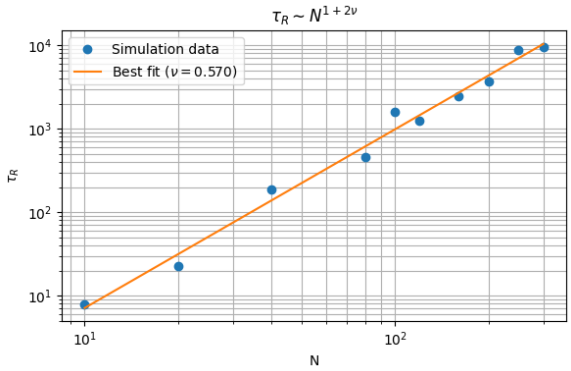
(c) Attempt 2.1



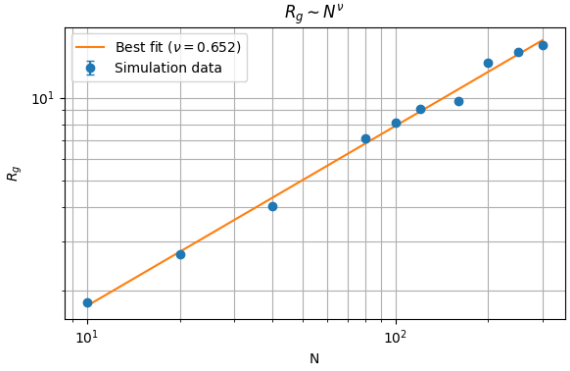
(d) Attempt 2.2



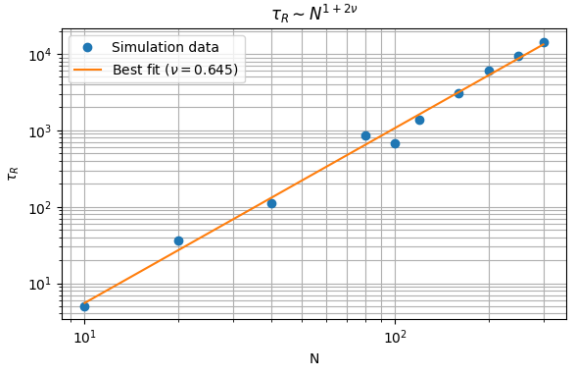
(a) Attempt 3.1



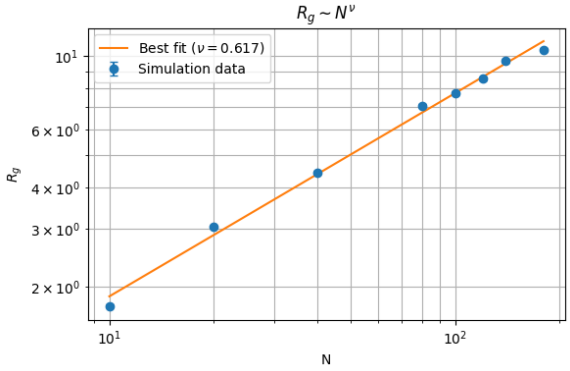
(b) Attempt 3.2



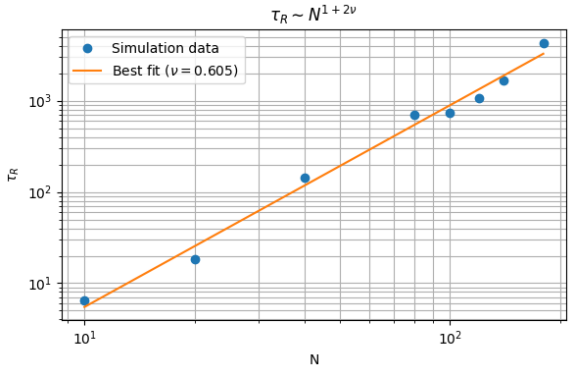
(c) Attempt 4.1



(d) Attempt 4.2



(e) Attempt 5.1



(f) Attempt 5.2

5 Parallel Averaging over Independent Runs

Within a run, successive R_g samples are strongly correlated because the polymer changes shape slowly. Hence, $\sqrt{\text{variance}/n}$ underestimates the true uncertainty. We therefore use independent runs with different random seeds; the spread between runs provides a more reliable error estimate. This also enables the polymer to explore more possible configurations and get a better estimate.

The computational cost grows rapidly with chain length: since $\tau_R \sim N^{2.2}$ and each step costs $\mathcal{O}(N)$, the cost of a run scales approximately as $N^{3.2}$. The simulations for different (N, run) pairs are independent, so they can be run simultaneously on different CPU cores. Thus, the 80 jobs (10 chain lengths $\times$ 8 runs) can be parallelized to reduce the total wall-clock time.

5.1 Algorithm parameters

For every (N, run) pair we use the following recipe.

- The chain is initialized as a self-avoiding random walk, with the first monomer at the box center and subsequent monomers placed at fixed bond length in random, non-overlapping directions.
- The run length is set from the estimated relaxation time τ_{est}, so that every N is treated equally: $3\,\tau_{\text{est}}$ of equilibration, then $20\,\tau_{\text{est}}$ of production. The equilibration was cut from 10 to 3 relaxation times because the R_g traces showed the chain relaxing within about one τ when it starts from a self-avoiding walk.
- In the production phase, R_g^2 and the first Rouse mode are sampled every $0.05\,\tau_{\text{est}}$, which gives 400 samples per run.
- Each run has a unique fixed random seed based on N and run number, which makes runs independent and results reproducible across threads.
- OpenMP framework is used for the parallelization with fortran.

The final value for each N is the mean $\bar{x}$ over the $M = 8$ runs, with standard error

$$\text{SEM} = \sqrt{\frac{\sum_{i=1}^{M}(x_i - \bar{x})^2}{M(M-1)}}. \tag{15}$$

This is applied to R_g and to τ_R.

6 Results and Discussion

We report two sets of runs. Set 1 covers $N = 10$ to 200 (10 sizes) and set 2 covers $N = 100$ to 300 (10 sizes). Each value is the mean of 8 runs. The overlap at $N = 100$ to 200 gives a check, and the values from the two sets agree within about two standard errors.

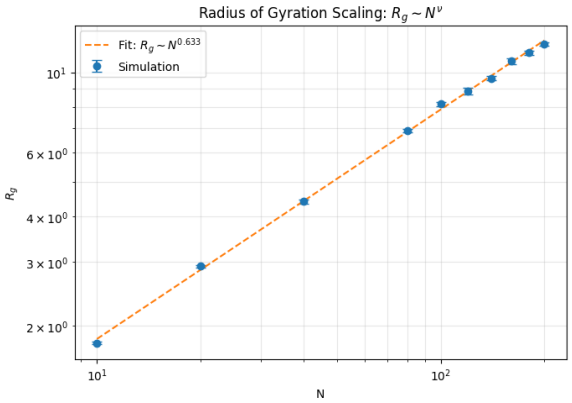
Set 1			Set 2		
N	R_g	τ_R	N	R_g	τ_R
10	1.795 ± 0.014	7.3 ± 0.4	100	8.04 ± 0.12	940 ± 86
20	2.919 ± 0.018	34.4 ± 2.3	120	8.85 ± 0.11	1522 ± 161
40	4.41 ± 0.06	131 ± 8	140	9.76 ± 0.13	2462 ± 210
80	6.91 ± 0.08	788 ± 90	160	10.47 ± 0.18	2446 ± 232
100	8.18 ± 0.12	1155 ± 95	180	11.22 ± 0.18	3736 ± 363
120	8.88 ± 0.21	1327 ± 152	200	12.02 ± 0.11	4298 ± 304
140	9.63 ± 0.13	2270 ± 267	220	13.02 ± 0.14	5577 ± 470
160	10.74 ± 0.19	3245 ± 351	240	13.27 ± 0.26	6452 ± 571
180	11.34 ± 0.15	3875 ± 344	260	13.91 ± 0.18	6876 ± 540
200	11.98 ± 0.14	4154 ± 231	300	15.66 ± 0.32	9396 ± 991

Table 3: Mean radius of gyration and Rouse relaxation time over 8 independent runs, with the standard error of the mean.

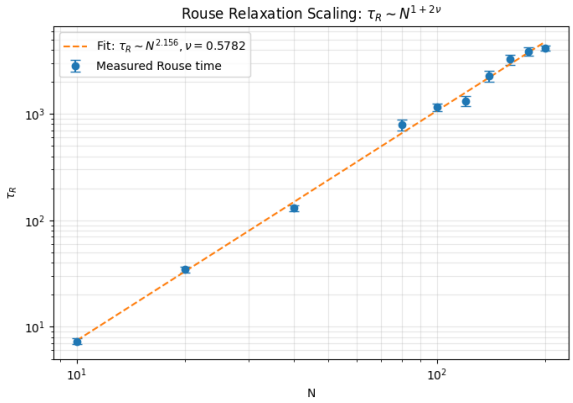
We fit straight lines to $\log R_g$ and $\log \tau_R$ against $\log N$. The fit gives the slope ν for R_g and the slope z for τ_R, from which $\nu = (z-1)/2$. The errors quoted are the standard errors of the fitted slope.

	N range	ν from R_g	z	ν from τ_R
Set 1	10 to 200	0.633 ± 0.007	2.16 ± 0.04	0.578 ± 0.020
Set 2	100 to 300	0.599 ± 0.012	2.05 ± 0.09	0.52 ± 0.04
Expected		0.588	2.176	0.588

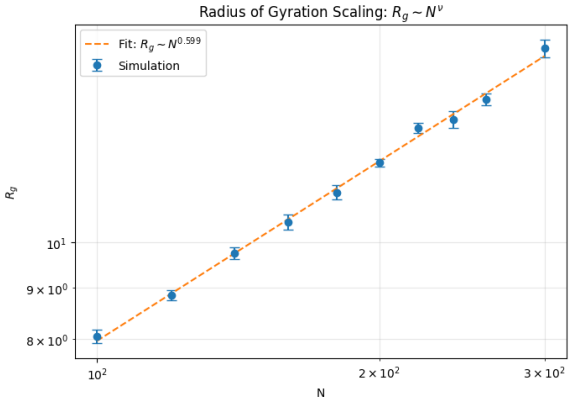
Table 4: Fitted exponents for the two sets of runs.



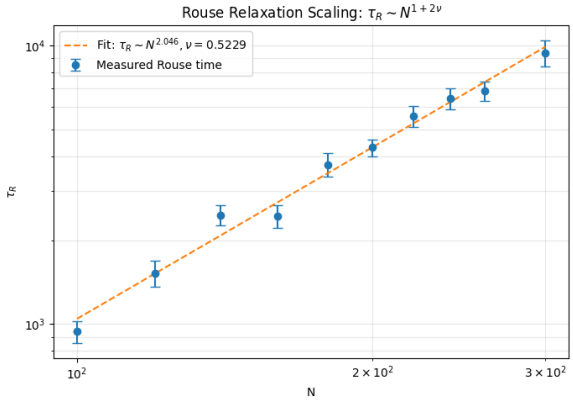
(a) Set 1: R_g against N ($N = 10$ to 200).



(b) Set 1: τ_R against N ($N = 10$ to 200).



(c) Set 2: R_g against N ($N = 100$ to 300).



(d) Set 2: τ_R against N ($N = 100$ to 300).

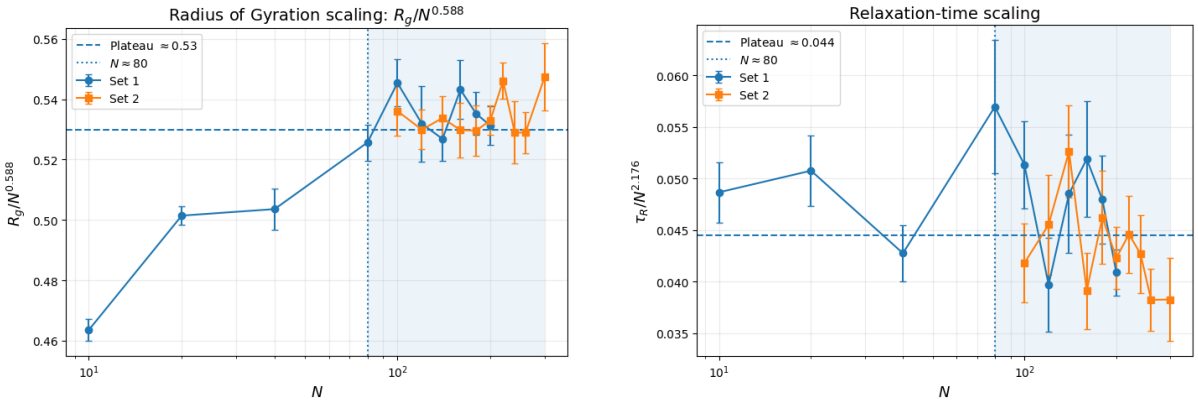
Figure 7: Scaling analysis for the two simulation sets. Solid lines are fits, while dashed lines show the expected slopes, placed to meet the corresponding fit at the middle of the N range.

Radius of gyration. For set 1 the fitted exponent is 0.633 ± 0.007, which is clearly above 0.588. For set 2, which has only longer chains, it is 0.599 ± 0.012. This differs from 0.588 by less than one standard error. So R_g agrees with the expected value when we work with longer polymers.

Relaxation time. For set 1 we find $z = 2.16 \pm 0.04$, so $\nu = 0.578 \pm 0.020$. This is consistent with $z = 2.176$. For set 2 we find $z = 2.05 \pm 0.09$ and $\nu = 0.52 \pm 0.04$. This is about 1.5 standard errors below the expected value, so the two are still compatible, but the central value is closer to the ideal-chain result $z = 2$ ($\nu = 0.5$). So the improved agreement for $N > 100$ is clear for R_g, but for τ_R the data is only consistent with theory but does not adhere to it very strongly.

6.1 Finite-size effects

The compensated quantity $R_g/N^{0.588}$ is shown in Figure 8a. For ideal asymptotic scaling, this ratio should approach a constant. The ratio increases from about 0.46 at $N = 10$ to approximately 0.53 by $N \sim 80$, after which it fluctuates around a roughly constant value. This indicates significant finite-size effects for short chains and suggests a crossover to the asymptotic regime at approximately $N \sim 80$–100. The exponent ν is therefore better estimated by averaging over larger N.



(a) Radius-of-gyration scaling

(b) Relaxation-time scaling

Figure 8: Compensated scaling analysis for the radius of gyration and Rouse relaxation time.

For the relaxation time, the corresponding compensated quantity is

$$\frac{\tau_R}{N^{1+2(0.588)}} = \frac{\tau_R}{N^{2.176}}.$$

As shown in Figure 8b, the data exhibit substantially larger fluctuations than the R_g scaling. Although the values remain of the same order at larger N, a clear plateau is not observed. This reflects the larger statistical uncertainty associated with estimating relaxation times from time-correlation data.

7 Ring Polymers

Section 6 dealt with linear chains. Here the last bead is also bonded to the first, so the chain closes into a ring. We measure R_g, the Rouse relaxation time τ_R and the mean squared displacements (MSD) for ring sizes $N = 100$ to 300. In a good solvent a ring is still a swollen self-avoiding object, so we expect the same exponents as for linear chains, $R_g \sim N^{\nu}$ and $\tau_R \sim N^{1+2\nu}$ with $\nu \approx 0.588$. Here we use $\delta t = 0.01$ similar to the time interval per iteration used in case of linear polymers.

7.1 Model and changes to the code

The potentials, parameters and BAOAB integrator are the same as in Table 1 and Section 4. Only four things differ from the linear-chain code.

- **Closing bond.** The harmonic spring is also applied between bead N and bead 1. The WCA repulsion still acts between all pairs.
- **Initial configuration.** The beads are placed on a circle in the xy plane with neighbouring beads exactly r_0 apart. The radius follows from the chord length, $2R\sin(\pi/N) = r_0$, so $R = r_0/[2\sin(\pi/N)]$. The closest non-bonded pair is then $2r_0\cos(\pi/N) > r_c$ apart, so nothing overlaps at the start. The simulation takes place in free space (no periodic box), so positions are never wrapped and can be used directly for the MSD.
- **Rouse mode.** A ring is periodic in the bead index, so its normal modes are Fourier modes instead of the cosine modes used for linear chains. The slowest mode, $p = 1$, is a degenerate cosine–sine pair,

$$\mathbf{X}_{1c} = \frac{1}{N}\sum_{i=1}^{N} \cos\left[\frac{2\pi(i-1)}{N}\right]\mathbf{r}_i, \qquad \mathbf{X}_{1s} = \frac{1}{N}\sum_{i=1}^{N} \sin\left[\frac{2\pi(i-1)}{N}\right]\mathbf{r}_i. \tag{16}$$

  Both parts relax with the same time. The autocorrelation function $C(t)$ is defined as before, with the dot product taken over all six components $(\mathbf{X}_{1c}, \mathbf{X}_{1s})$, and τ_R is the integrated autocorrelation time with the same cut at the first $C \le 0$.
- **Planning time.** The wavelength of the slowest ring mode is half that of the linear $p = 1$ mode. The ring $p = 1$ mode therefore relaxes like the linear $p = 2$ mode, and the relaxation time is a quarter of the linear value,

$$\tau_{\text{est}}^{\text{ring}} = \frac{\gamma\, N^{1+2\nu}}{12\pi^2} = \frac{1}{4}\,\tau_{\text{est}}^{\text{linear}}. \tag{17}$$

7.2 Run parameters and parallelisation

Run lengths are set from $\tau_{\text{est}}^{\text{ring}}$ in the same way as for the linear chains, so that every N is treated equally.

- $10\,\tau_{\text{est}}$ of equilibration, then $20\,\tau_{\text{est}}$ of production. The equilibration is longer than for the linear chains ($3\,\tau_{\text{est}}$) because the ring starts from a perfect circle, which is far from a typical swollen ring, and so it needs time to relax.
- R_g^2 and the Rouse mode are sampled every $0.05\,\tau_{\text{est}}$ during production, which gives about 400 samples per run.

- The MSD uses a fixed absolute sampling interval of 0.1 time units (not a multiple of τ), so the short-time behaviour is resolved for every N. Lag times are log-spaced with 20 points per decade, and all available time origins are used.
- The MSD is measured for five beads equally spaced along the contour, $i_k = 1 + (k-1)N/5$, together with the centre of mass.

The 16 ring sizes are independent jobs. They are distributed over 8 OpenMP threads with `schedule(dynamic,1)`, and the queue is ordered from the largest N to the smallest. Since the cost grows roughly as $N^{3.2}$, the long runs start first and the short ones fill the gaps at the end. Each job runs on a single thread with its own random-number state (including a thread-private Box–Muller spare), and the seed depends only on N. The results are therefore reproducible and independent of the number of threads. Unlike the linear chains, there is *one* run per N here and not eight, so the errors quoted below are weaker than in Section 6.

7.3 Mean squared displacements

For the tracked beads i we compute three quantities at lag time t, averaged over time origins t_0:

$$g_1(t) = \left\langle |\mathbf{r}_i(t_0+t) - \mathbf{r}_i(t_0)|^2 \right\rangle, \tag{18}$$

$$g_2(t) = \left\langle |[\mathbf{r}_i - \mathbf{r}_{\mathrm{cm}}](t_0+t) - [\mathbf{r}_i - \mathbf{r}_{\mathrm{cm}}](t_0)|^2 \right\rangle, \tag{19}$$

$$g_3(t) = \left\langle |\mathbf{r}_{\mathrm{cm}}(t_0+t) - \mathbf{r}_{\mathrm{cm}}(t_0)|^2 \right\rangle. \tag{20}$$

Here g_1 is the motion of a bead in the lab frame, g_2 is the motion of a bead relative to the centre of mass (internal motion of the ring), and g_3 is the motion of the centre of mass. g_1 and g_2 are averaged over the five tracked beads.

For a free-draining chain with excluded volume and no hydrodynamics, the expected behaviour at times between the monomer time and τ_R is

$$g_1,\, g_2 \sim t^{\,2\nu/(1+2\nu)} \approx t^{0.54}, \qquad g_3 \sim t^1, \tag{21}$$

where the ideal Rouse value of the first exponent is 0.5. The centre of mass diffuses freely, so g_3 should be linear in t at times beyond the friction time $m/\gamma = 1$.

Results for rings

7.4 Radius of gyration

Table 5 lists R_g for every ring size, with the error from the within-run samples. Since successive samples are correlated, this error is only a lower bound (see Section on parallel averaging). A power-law fit gives

$$\nu_{\mathrm{ring}} = 0.589 \pm 0.022, \qquad R^2 = 0.980, \tag{22}$$

which is in very good agreement with 0.588.

The compensated quantity $R_g/N^{0.588}$ in Table 5 is flat at about 0.39–0.40 over the whole range, with no systematic trend. Unlike the linear chains (Figure 8a), there is no finite-size rise at small N. This is partly because the smallest ring here is already $N = 100$, which for the linear chains lies in the asymptotic regime. The scatter of a few percent is larger than the quoted errors, which again shows that the within-run errors are too small.

At the same N, the ring is more compact than the linear chain. The ratio $R_g^{\text{ring}}/R_g^{\text{linear}}$ lies between 0.70 and 0.76 (comparing with the averaged values in Table 3), which is close to the Gaussian value $\sqrt{1/2} \approx 0.71$. This is a comparison between a single ring run and eight-run linear averages, so we treat it only as a rough check.

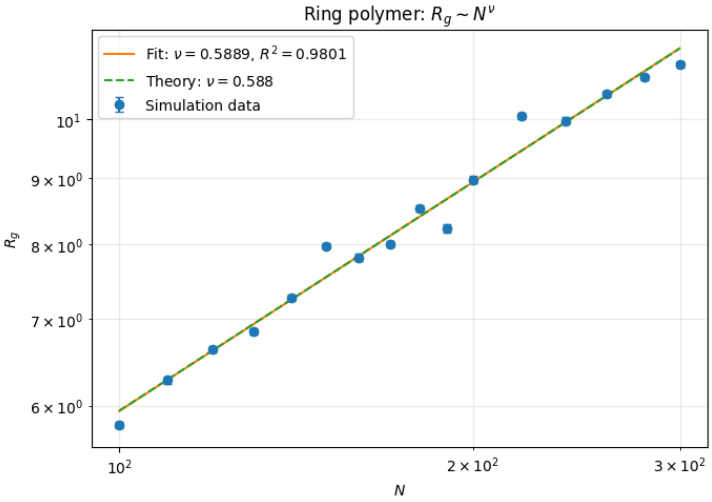


Figure 9: Ring polymers: R_g against N on log–log axes. The solid line is the fit ($\nu = 0.589$) and the dashed line has the expected slope 0.588, placed at the geometric mean of the data.

N	R_g	$R_g/N^{0.588}$	τ_{est}	τ_R (measured)
100	5.80 ± 0.03	0.386	190	215
110	6.28 ± 0.04	0.396	234	401
120	6.64 ± 0.04	0.398	282	317
130	6.85 ± 0.04	0.392	336	557
140	7.27 ± 0.05	0.398	395	455
150	7.98 ± 0.04	0.419	459	831
160	7.81 ± 0.05	0.395	528	708
170	8.00 ± 0.04	0.390	603	1111
180	8.53 ± 0.05	0.403	682	1260
190	8.23 ± 0.06	0.376	768	963
200	8.98 ± 0.06	0.398	858	1191
220	10.06 ± 0.06	0.422	1056	1600
240	9.97 ± 0.06	0.397	1276	1743
260	10.46 ± 0.06	0.398	1519	2410
280	10.77 ± 0.06	0.392	1785	2614
300	11.03 ± 0.06	0.386	2074	3125

Table 5: Ring polymer results from a single run per N. The error on R_g assumes independent samples and is therefore a lower bound. τ_{est} is the planning estimate $\gamma N^{1+2\nu}/(12\pi^2)$ and τ_R is the measured integrated autocorrelation time of the $p = 1$ ring mode.

7.5 Relaxation time

The fit to the measured relaxation times gives

$$z_{\text{ring}} = 2.28 \pm 0.13, \qquad \nu = \frac{z-1}{2} = 0.64 \pm 0.07, \qquad R^2 = 0.955. \tag{23}$$

This is above the expected $z = 2.176$ ($\nu = 0.588$), but only by about 0.8 standard errors, so the two are compatible. The fit is noticeably worse than for R_g. The measured τ_R is not even monotonic in N: for example $\tau_R = 401$ at $N = 110$ and 317 at $N = 120$, and 1260 at $N = 180$ against 963 at $N = 190$. This is the same noise problem as for the linear chains. τ_R comes from a single, noisy autocorrelation function, and the cut at the first $C \leq 0$ makes the result sensitive to which noise fluctuation crosses zero first. Taking the quoted exponent at face value would therefore overstate our confidence.

The measured values are 1.1 to 1.9 times the planning estimate τ_{est} (mean about 1.5). This is not a concern, as τ_{est} is the free Rouse prefactor, which is only used to set the run length. What matters is that all runs last many measured relaxation times. The measured ring times are about 0.2 to 0.35 of the linear-chain times at the same N (Tables 3 and 5), which is consistent in order of magnitude with the factor 1/4 expected from the mode argument above.

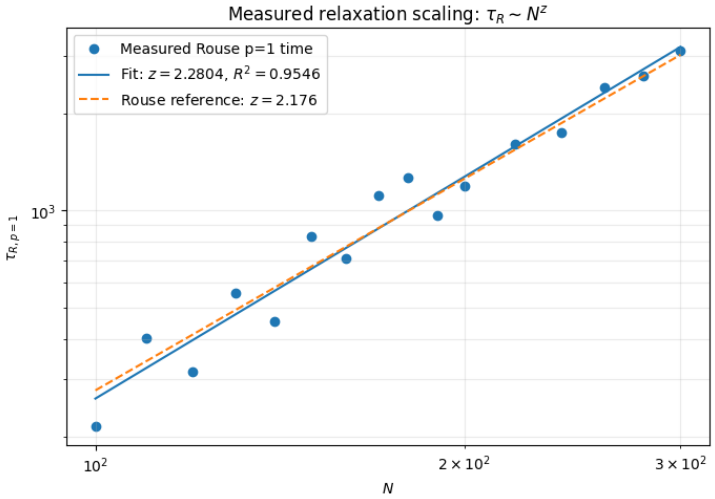


Figure 10: Ring polymers: measured $p = 1$ Rouse relaxation time against N. The solid line is the fit ($z = 2.28$) and the dashed line has the Rouse slope $1 + 2\nu = 2.176$.

7.6 MSD curves

Figure 11 shows g_1, g_2 and g_3 for all 16 ring sizes. The bead MSDs g_1 and g_2 are not measured for every bead. For each ring we track only five beads, equally spaced along the contour, and average over them and over all time origins. The centre-of-mass MSD g_3 needs no bead selection, since it uses the centre of mass of the whole ring. A ring has no chain ends, so all beads are statistically equivalent, and no tracked bead is special as an end bead would be in a linear chain. Averaging over five beads therefore only improves the statistics. The five beads are not independent, since neighbouring beads move together on scales below the ring size. The averaging thus reduces the noise less than a factor of $\sqrt{5}$ would suggest. It also means we store only the positions of five beads and the centre of mass rather than the full trajectory, which keeps the output files small. At short times ($t \lesssim 1$) the motion is ballistic and all sizes coincide, because a bead has not yet felt the rest of the ring. At later times the curves of different N remain close to each other, since the intermediate regime is controlled by the local dynamics of the chain and not by N. The size dependence appears only near the end of each curve, where the g_2 curves flatten at a plateau set by the size of the ring (of order R_g^2) once $t \gtrsim \tau_R$. At large lag times there are few independent time origins, so the tails of all three curves are noisy.

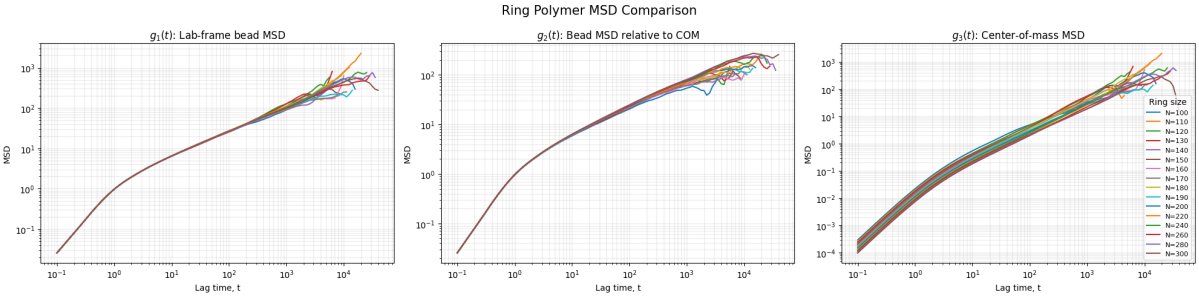


Figure 11: MSDs of the ring polymers for $N = 100$ to 300: (left) lab-frame bead MSD g_1, (centre) bead MSD relative to the centre of mass g_2, (right) centre-of-mass MSD g_3.

7.7 MSD scaling exponents

We fit $g_k \sim t^{\alpha_k}$ in the window $1 \le t \le 1000$ (61 log-spaced points) for each N. The results for the three quantities are summarised in Table 6. The fits to individual curves have $R^2 > 0.985$ throughout.

N	α_1 (bead, lab frame)	α_2 (bead, CM frame)	α_3 (centre of mass)
100	0.623	0.565	1.005
110	0.654	0.592	1.072
120	0.668	0.603	1.103
130	0.663	0.591	1.118
140	0.639	0.588	1.073
150	0.650	0.588	1.109
160	0.639	0.580	1.105
170	0.639	0.589	1.097
180	0.654	0.603	1.127
190	0.648	0.609	1.079
200	0.639	0.599	1.092
220	0.637	0.600	1.089
240	0.648	0.614	1.114
260	0.640	0.611	1.088
280	0.641	0.617	1.085
300	0.645	0.618	1.097
Mean $\pm$ std. dev.	0.645 ± 0.011	0.598 ± 0.014	1.091 ± 0.028
Expected	0.54	0.54	1

Table 6: Fitted MSD exponents $g_k \sim t^{\alpha_k}$ for each ring size, in the window $1 \le t \le 1000$ (61 points). The log–log fits have $R^2 > 0.985$ in every case. The mean and standard deviation are taken over the 16 ring sizes, and the expected values are $2\nu/(1+2\nu) \approx 0.54$ for the internal motion and 1 for diffusion of the centre of mass.

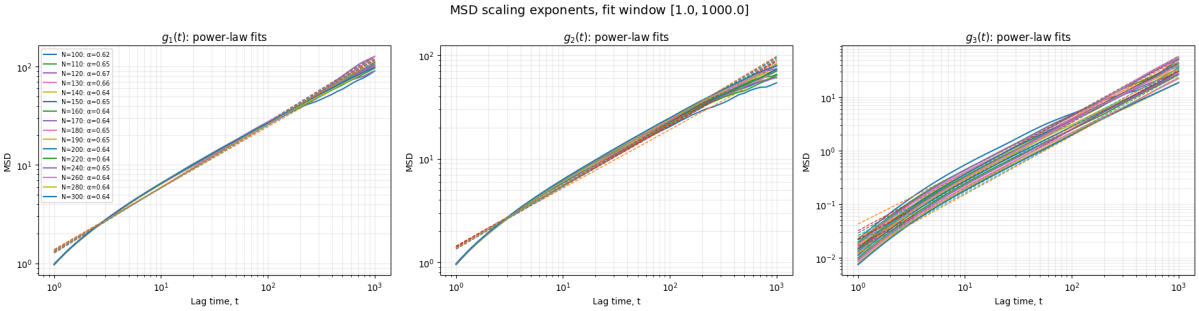


Figure 12: Power-law fits (dashed) to g_1, g_2 and g_3 in the window $1 \leq t \leq 1000$.

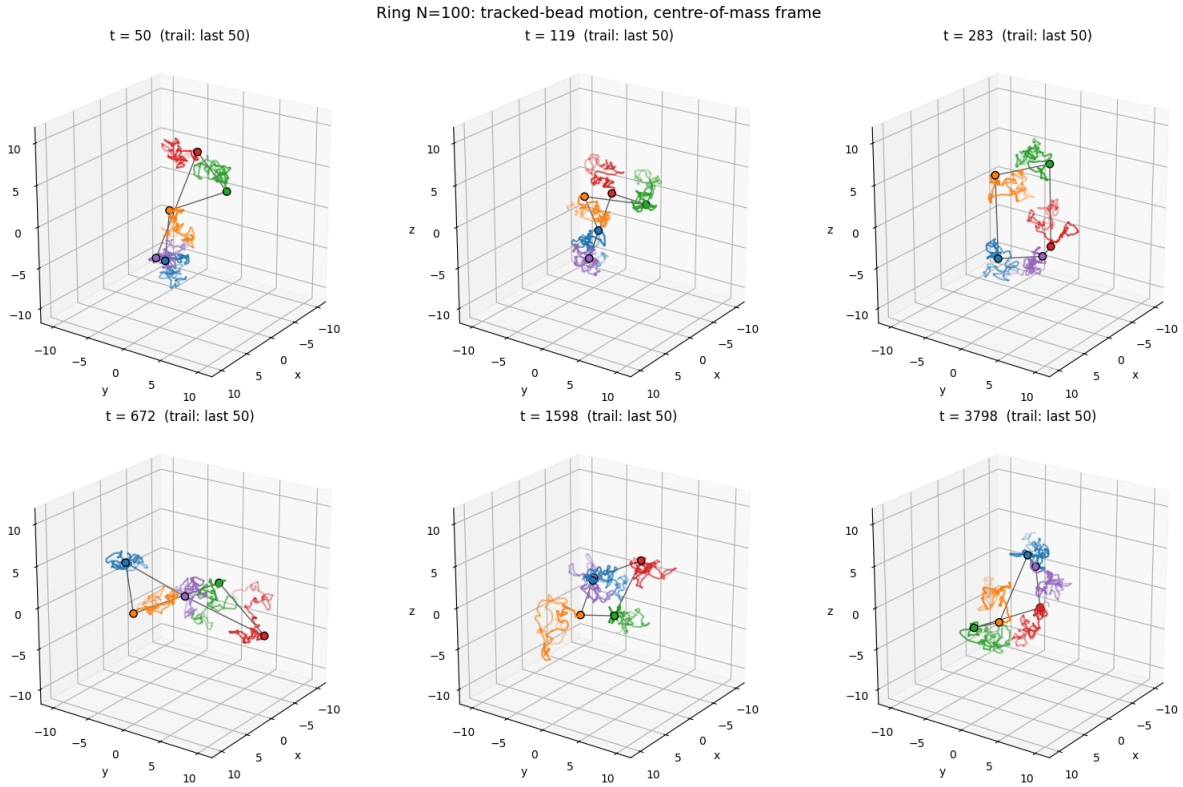


Figure 13: Snapshot of particle trajectories for particles : 1, 21, 41, 61, 81 over simulation time, the tails represent monomeric trajectory over a 50 iteration window.

The exponents show the following.

- **Centre of mass.** $\alpha_3 \approx 1.09$ is close to the diffusive value 1 but is slightly above it for every N. The window starts at $t = 1 = m/\gamma$, so it includes the end of the crossover from ballistic motion ($\alpha = 2$) to diffusion, which pushes the fitted exponent above 1. I have not performed furtehr tests to verify which is the ideal window for such readings.
- **Internal motion.** $\alpha_2 \approx 0.60$ lies between the ideal Rouse value 0.5 and the excluded-volume value 0.54, and slightly above both. The same crossover from the short-time regime probably biases it upward.

- **Lab frame.** $\alpha_1 \approx 0.65$ is higher than α_2. This is expected: the lab-frame displacement contains the internal motion plus the centre-of-mass diffusion, $g_1 \approx g_2 + g_3$ (the cross term averages out), so the linear g_3 contribution raises the effective exponent. It follows that α_1 is not a pure measurement of the internal-mode exponent, and α_2 is the better quantity to compare with theory.

Both fit windows are of similar length for all sizes, and since τ_R ranges from about 200 to 3000, the upper end $t = 1000$ lies beyond τ_R for the smallest rings and well below it for the largest. The absence of a visible trend in α with N suggests the result is not very sensitive to this, but a window scaled with τ_R would be a cleaner test.

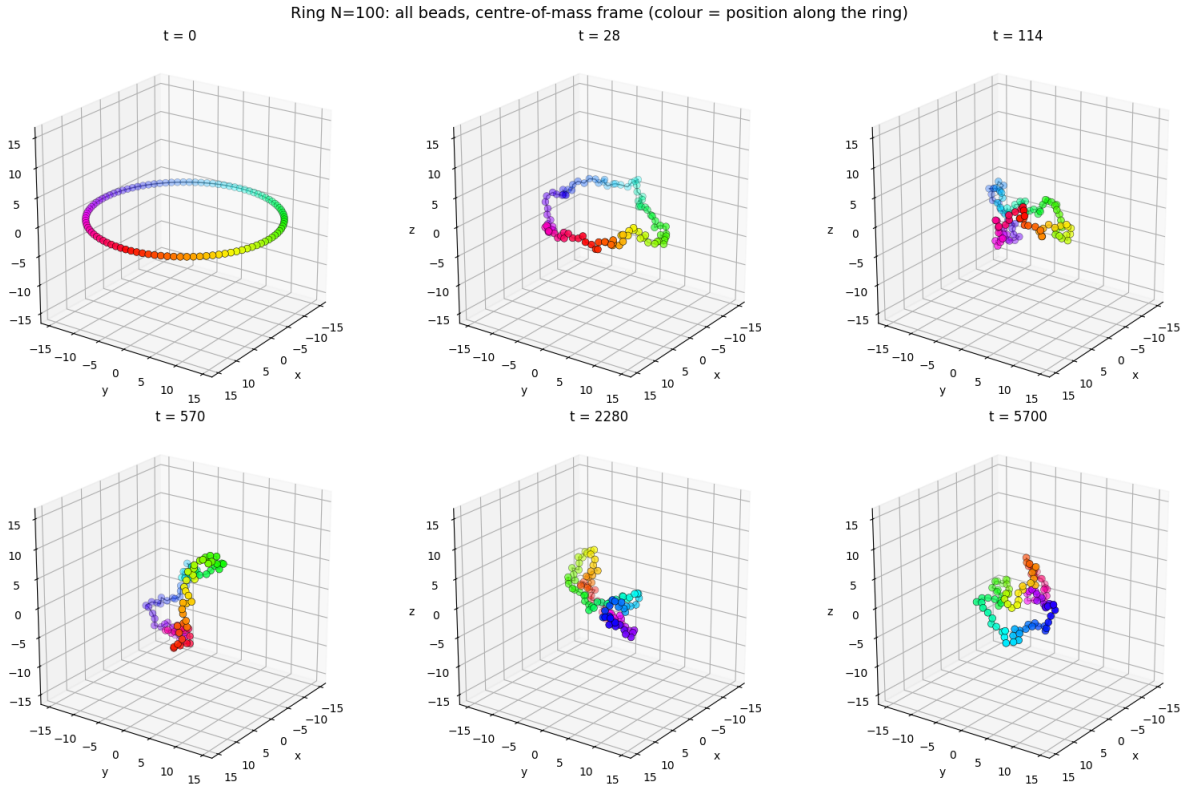


Figure 14: Ring polymer evolution over $T = 5700$ and $\delta t = 0.001$

7.8 Summary and limitations

	Fitted	Expected
ν from R_g	0.589 ± 0.022	0.588
z (from τ_R)	2.28 ± 0.13	2.176
ν from τ_R	0.64 ± 0.07	0.588
MSD α_2 (internal)	≈ 0.60	0.54
MSD α_3 (CM)	≈ 1.09	1

Table 7: Summary of the exponents for ring polymers ($N = 100$ to 300).

The ring R_g gives the Flory exponent very accurately, and the centre of mass diffuses normally. The ring relaxation time is consistent with $N^{1+2\nu}$ but noisy, in the same way as for the linear chains. The main limitation of this section is that each N was simulated only once. The error

bars on R_g are therefore lower bounds, there is no error bar at all on τ_R or on the MSD exponents, and the scatter in τ_R against N shows that single-run values are not reliable at the level of a few tens of percent. Repeating each size with several seeds, as was done for the linear chains in Section 6, would give proper standard errors and should reduce the noise in z.
